

Observed-Based Adaptive Fuzzy Decentralized Tracking Control for Switched Uncertain Nonlinear Large-Scale Systems With Dead Zones

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Abstract—In this paper, the problem of adaptive fuzzy decentralized output-feedback control design is investigated for a class of switched nonlinear large-scale systems in strict-feedback form. The considered nonlinear large-scale systems contain the unknown nonlinearities and dead zones, the switching signals with average dwell time, and without the direct requirement of the states being available for feedback. Fuzzy logic systems are utilized to approximate the unknown nonlinear functions, a fuzzy switched decentralized state observer is designed and thus via it the immeasurable states are obtained. By applying the adaptive decentralized backstepping design technique, an adaptive fuzzy decentralized output-feedback tracking control approach is developed for the switched subsystems. The stability of the whole closed-loop system is proved by using the Lyapunov function and the average dwell-time methods. Satisfactory tracking performance is achieved under the switching signals with average dwell time. The simulation example is provided to indicate the effectiveness of the proposed control method.

Index Terms—Average dwell time, fuzzy adaptive decentralized control, fuzzy logic systems (FLSs), switched nonlinear large-scale systems.

I. INTRODUCTION

MANY physical systems such as power industry, aerospace system, chemical engineering system and telecommunication network are composed of interconnections of lower-dimensional subsystems [1]–[4]. Unique technical issues naturally arise from control of the large-scale systems to achieve the corresponding control goals under subsystem interactions. Meanwhile, a large-scale system usually possesses nonlinear and uncertain characteristics which results in control design difficulty [5], [6]. It is well known that the decentralized control structure naturally alleviates the computational burden associated with a centralized control scheme. Hence, decentralized robust control as an effective control approach has been extensively studied for uncertain nonlinear

large-scale systems [7]. By far, there has been an increasing interest in the development of a decentralized control design method and theory for nonlinear large-scale systems. In [8]–[10], adaptive neural network and fuzzy decentralized state feedback control designs were proposed for large-scale nonlinear systems in strict feedback form. In [11]–[14], adaptive neural network and fuzzy decentralized output feedback control methods were studied for large-scale nonlinear systems by designing a state observer for estimating the unmeasured states. Adaptive neural network decentralized robust stabilization was investigated in [15] for uncertain stochastic interconnected nonlinear systems. The above mentioned adaptive fuzzy or neural decentralized control approaches can provide an effective methodology to control those uncertain nonlinear large-scale systems. Nevertheless, they need not require that the nonlinear functions included in the controlled systems be known or be linearly parameterized. However, the above mentioned results are only suitable for the non-switched nonlinear large-scale systems, instead of the switched nonlinear large-scale systems.

It is well known that switched systems are an important class of hybrid systems, which can be described by a family of subsystems and a rule that orchestrates the switching between them. In fact, many real-world systems, such as mechanical systems, power electronic, automotive industry, air traffic control, etc., can be expressed as switched systems. Thus the control design and stability on the switched systems have attracted the great attention of many scholars [16]–[21].

In recent years, by applying the backstepping control design technique, several adaptive control methods have been proposed for some classes of switched nonlinear systems [22]–[29]. Among them, two state feedback control approaches in [22]–[24] have been investigated based on the common Lyapunov function method for a class of switched nonlinear systems, but the nonlinear functions of controlled systems in [22]–[24] are required to be known functions. In [25], an adaptive neural network stabilizer and a switching law have been designed for a class of switched nonlinear systems with a triangular structure, and neural networks are utilized to approximate the unknown nonlinear functions of controlled systems. Two adaptive neural control schemes have been proposed for a class of switched nonlinear systems in [26] and [27], and an admissible switching signal with

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an average dwell-time technique has been given. In [28], an adaptive neural network feedback control scheme of nonlinear switched impulsive systems has been given under an all admissible switched strategy. The neural networks are used to compensate for the nonlinear uncertainties of switched impulsive systems. In [29], an adaptive neural network control method has been presented for a class of switched nonlinear systems with switching jumps and uncertainties based on dwell-time property. However, the aforementioned results are all restricted to the uncertain nonlinear systems and it is assumed that the states are available for measurement. Therefore, they cannot be used to control the switched nonlinear systems with immeasurable states. In addition, the switched plants under study in the above references are single-input and single-output switched nonlinear systems. To the best of our knowledge, to date, there are no results on adaptive fuzzy decentralized output feedback control available for uncertain switched nonlinear large-scale systems with immeasurable states.

This paper investigates the adaptive fuzzy decentralized control design problem for a class of switched nonlinear large-scale systems with unknown nonlinear functions, dead zones, and unmeasurable states. In the control design, fuzzy logic systems (FLSs) are first utilized to approximate the unknown nonlinear functions, and a fuzzy switched decentralized state observer is designed and thus via it the immeasurable states are obtained. By adaptive backstepping design technique, an adaptive fuzzy decentralized output-feedback tracking control approach is developed. The stability of the whole closed-loop system is proved based on Lyapunov function method and average dwell-time method. A satisfactory tracking performance is achieved under the switching signals with the average dwell time. Compared with the existed literatures, the main contributions of this paper are as follows.

- 1) This paper first proposed an adaptive fuzzy decentralized tracking output feedback control method for a class of switched nonlinear large-scale systems in strict-feedback form. The proposed adaptive control method has solved the state unmeasured problem via designing fuzzy switched state observer. Although [25]–[29] also addressed the adaptive fuzzy or neural control design problem for switched nonlinear strict-feedback systems, they all required that the states must be measurable.
- 2) This paper first investigated the adaptive fuzzy decentralized output-feedback control design problem for the uncertain switched nonlinear large-scale systems. The proposed control scheme cannot only guarantee the stability of the whole switched control system, but also can lead to a satisfactory tracking control performance. Note that the previous results in [8]–[15] only focused on the nonswitched nonlinear large-scale systems, thus they cannot be applied to the switched problems in this paper.
- 3) The results in this paper are developed with full consideration of the dead-zone problem. By constructing a dead-zone compensator, the proposed adaptive fuzzy decentralized output-feedback control approach can overcome the unknown dead-zone problem of uncertain switched nonlinear large-scale systems.

The remainder of this paper is organized as follows. The problem formulation and preliminaries are described in Section II. The switched fuzzy decentralized state observer design is given in Section III. The adaptive fuzzy output-feedback control scheme and stability analysis are present in Section IV. The simulation example is given in Section V, and followed by Section VI which concludes this paper.

II. PROBLEM FORMULATIONS AND PRELIMINARIES

A. System Descriptions and Assumptions

Consider a class of uncertain switched nonlinear large-scale systems that is composed of N subsystems interconnected by their outputs. The i th switched subsystem can be described by

$$\begin{cases} \dot{x}_{i,1} = x_{i,2} + f_{i,1}^{\sigma(t)}(x_{i,1}) + \Delta_{i,1}^{\sigma(t)}(\bar{y}) \\ \dot{x}_{i,2} = x_{i,3} + f_{i,2}^{\sigma(t)}(x_{i,2}) + \Delta_{i,2}^{\sigma(t)}(\bar{y}) \\ \vdots \\ \dot{x}_{i,n_i-1} = x_{i,n_i} + f_{i,n_i-1}^{\sigma(t)}(x_{i,n_i-1}) + \Delta_{i,n_i-1}^{\sigma(t)}(\bar{y}) \\ \dot{x}_{i,n_i} = D_i^{\sigma(t)}(u_i^{\sigma(t)}) + f_{i,n_i}^{\sigma(t)}(x_{i,n_i}) + \Delta_{i,n_i}^{\sigma(t)}(\bar{y}) \\ y_i = x_{i,1} \end{cases} \quad (1)$$

where $x_{i,j} = [x_{i,1}, x_{i,2}, \dots, x_{i,j}]^T \in \mathbb{R}^i$, $i = 1, 2, \dots, N$; $j = 1, 2, \dots, n_i$ are the states and $y_i \in \mathbb{R}$ is the output $\bar{y} = [y_1, y_2, \dots, y_N]$. The function $\sigma(t) : [0, \infty) \rightarrow M = \{1, 2, \dots, m\}$ is switching signal which is assumed to be a piecewise continuous (from the right) function of time. Moreover, $\sigma(t) = k$ implies that the k th switched large-scale system is active. $f_{i,j}^{\sigma(t)}(x_{i,j})$, $i = 1, 2, \dots, N$; $j = 1, 2, \dots, n_i$, are unknown smooth nonlinear functions. $\Delta_{i,j}^{\sigma(t)}(\bar{y})$ are unknown smooth functions representing the interconnection effects between the i th subsystem and the other subsystems. $D_i^{\sigma(t)}(u_i^{\sigma(t)}) \in \mathbb{R}$ is the output of the dead zone. In addition, we assume that the state of the system (1) does not jump at the switching instants, i.e., the solution is everywhere continuous, which is a standard assumption in the switched system literature [18]–[28].

Similar to [30] and [31], the output of the dead zone $D_i^{\sigma(t)}(u_i^{\sigma(t)})$ is described by

$$D_i^{\sigma(t)}(u_i^{\sigma(t)}) \triangleq \begin{cases} m_{i,r}^{\sigma(t)}(u_i^{\sigma(t)} - d_{i,r}^{\sigma(t)}) & \text{if } u_i^{\sigma(t)} \geq d_{i,r}^{\sigma(t)} \\ 0 & \text{if } -d_{i,l}^{\sigma(t)} < u_i^{\sigma(t)} < d_{i,r}^{\sigma(t)} \\ m_{i,l}^{\sigma(t)}(u_i^{\sigma(t)} + d_{i,l}^{\sigma(t)}) & \text{if } u_i^{\sigma(t)} \leq -d_{i,l}^{\sigma(t)}. \end{cases} \quad (2)$$

In (2), $u_i^{\sigma(t)} \in \mathbb{R}$ is the input to the dead zone; $m_{i,r}^{\sigma(t)}$ and $m_{i,l}^{\sigma(t)}$ denote the slope of the dead zone and $d_{i,r}^{\sigma(t)}$ and $d_{i,l}^{\sigma(t)}$ represent the dead zone width parameters. The assumptions for the controlled system and dead zone are the following. The dead-zone outputs $D_i^{\sigma(t)}(u_i^{\sigma(t)})$ are not available for measurement. The dead-zone parameters $m_{i,r}^{\sigma(t)}$, $m_{i,l}^{\sigma(t)}$, $d_{i,r}^{\sigma(t)}$, and $d_{i,l}^{\sigma(t)}$ are unknown, but their signs are known ($m_{i,r}^{\sigma(t)} > 0$, $m_{i,l}^{\sigma(t)} > 0$, $d_{i,r}^{\sigma(t)} \geq 0$, and $d_{i,l}^{\sigma(t)} \leq 0$, respectively). Dead-zone slopes

are bounded by known constants $m_{i,r \min}^{\sigma(t)}$, $m_{i,r \max}^{\sigma(t)}$, $m_{i,l \min}^{\sigma(t)}$, and $m_{i,l \max}^{\sigma(t)}$ such that $0 < m_{i,r \min}^{\sigma(t)} \leq m_{i,r}^{\sigma(t)} \leq m_{i,r \max}^{\sigma(t)}$ and $0 < m_{i,l \min}^{\sigma(t)} \leq m_{i,l}^{\sigma(t)} \leq m_{i,l \max}^{\sigma(t)}$. The dead-zone inverse technique is useful for compensating the dead-zone effect [31]. Setting $u_{i,d}^{\sigma(t)}$ as the control input from the controller to achieve the control objective for the plant without a dead zone, the following control signal $u_i^{\sigma(t)}$ is generated according to the certainty equivalence dead-zone inverse:

$$u_i^{\sigma(t)} = \left(D_i^{\sigma(t)}(u_{i,d}^{\sigma(t)}) \right)^{-1} = \frac{u_{i,d}^{\sigma(t)} + \hat{d}_{i,r,m}^{\sigma(t)} \delta_i^{\sigma(t)}}{\hat{m}_{i,r}^{\sigma(t)}} \delta_i^{\sigma(t)} + \frac{u_{i,d}^{\sigma(t)} + \hat{d}_{i,l,m}^{\sigma(t)} (1 - \delta_i^{\sigma(t)})}{\hat{m}_{i,l}^{\sigma(t)}} \quad (3)$$

where $\hat{m}_{i,r}^{\sigma(t)}$, $\hat{m}_{i,l}^{\sigma(t)}$, $\hat{d}_{i,r,m}^{\sigma(t)}$, and $\hat{d}_{i,l,m}^{\sigma(t)}$ are estimations of $m_{i,r}^{\sigma(t)}$, $m_{i,l}^{\sigma(t)}$, $d_{i,r}^{\sigma(t)}$, and $d_{i,l}^{\sigma(t)}$, respectively, and

$$\delta_i^{\sigma(t)} = \begin{cases} 1 & \text{if } u_{i,d}^{\sigma(t)} \geq 0 \\ 0 & \text{if } u_{i,d}^{\sigma(t)} < 0. \end{cases} \quad (4)$$

The resulting errors between $u_i^{\sigma(t)}$ and $u_{i,d}^{\sigma(t)}$ are given by

$$D_i^{\sigma(t)}(u_i^{\sigma(t)}) - u_{i,d}^{\sigma(t)} = \left(\tilde{d}_{i,r,m}^{\sigma(t)} - \frac{u_{i,d}^{\sigma(t)} + \hat{d}_{i,r,m}^{\sigma(t)} \tilde{m}_{i,r}^{\sigma(t)}}{\hat{m}_{i,r}^{\sigma(t)}} \right) \delta_i^{\sigma(t)} + \left(\tilde{d}_{i,l,m}^{\sigma(t)} - \frac{u_{i,d}^{\sigma(t)} + \hat{d}_{i,l,m}^{\sigma(t)} \tilde{m}_{i,l}^{\sigma(t)}}{\hat{m}_{i,l}^{\sigma(t)}} \right) \times (1 - \delta_i^{\sigma(t)}) + \varepsilon_{i,d}^{\sigma(t)} \quad (5)$$

where $\tilde{m}_{i,r}^{\sigma(t)} = \hat{m}_{i,r}^{\sigma(t)} - m_{i,r}^{\sigma(t)}$, $\tilde{m}_{i,l}^{\sigma(t)} = \hat{m}_{i,l}^{\sigma(t)} - m_{i,l}^{\sigma(t)}$, $\tilde{d}_{i,r,m}^{\sigma(t)} = \hat{d}_{i,r,m}^{\sigma(t)} - d_{i,r,m}^{\sigma(t)}$, and $\tilde{d}_{i,l,m}^{\sigma(t)} = \hat{d}_{i,l,m}^{\sigma(t)} - d_{i,l,m}^{\sigma(t)}$ are parameters errors. $\varepsilon_{i,d}^{\sigma(t)}$ can be expressed as $\varepsilon_{i,d}^{\sigma(t)} = -m_{i,r}^{\sigma(t)} \chi_{i,r}^{\sigma(t)} (u_i^{\sigma(t)} - d_{i,r}^{\sigma(t)}) - m_{i,l}^{\sigma(t)} \chi_{i,l}^{\sigma(t)} (u_i^{\sigma(t)} - d_{i,l}^{\sigma(t)})$, and $\varepsilon_{i,d}^{\sigma(t)}$ is bounded, where

$$\chi_{i,r}^{\sigma(t)} = \begin{cases} 1, & \text{if } 0 \leq u_i^{\sigma(t)} < d_{i,r}^{\sigma(t)} \\ 0, & \text{otherwise} \end{cases}$$

and

$$\chi_{i,l}^{\sigma(t)} = \begin{cases} 1, & \text{if } d_{i,l}^{\sigma(t)} < u_i^{\sigma(t)} < 0 \\ 0, & \text{otherwise.} \end{cases}$$

Assumption 1: For the nonlinear functions $\Delta_{i,j}^{\sigma(t)}(\bar{y})$ satisfy

$$\left| \Delta_{i,j}^{\sigma(t)}(\bar{y}) \right| \leq \sum_{l=1}^N q_{i,l}^{\sigma(t)} \beta_l^{\sigma(t)}(|y_l|) \quad i = 1, 2, \dots, N; j = 1, 2, \dots, n_i \quad (6)$$

where $q_{i,l}^{\sigma(t)}$ is an unknown constant denoting the strength of the interaction; $\beta_l^{\sigma(t)}(|y_l|) \geq 0$ is a known nonlinear smooth function. Thus, according to [5], [7], and [15], there exists smooth nonnegative function $\bar{h}_{i,l}^{\sigma(t)}(y_l)$ such that

$$\left(\sum_{l=1}^N q_{i,l}^{\sigma(t)} \beta_l^{\sigma(t)}(|y_l|) \right)^2 \leq \sum_{l=1}^N \bar{h}_{i,l}^{\sigma(t)} y_l^2 + 2 \left(\sum_{l=1}^N \beta_l^{\sigma(t)}(0) \right)^2. \quad (7)$$

Remark 1: It should be mentioned that if for $t \in [0, \infty)$, the switching signal $\sigma(t) = k$, then the system (1) represents a class of nonswitched nonlinear large-scale systems in strict-feedback form, which is investigated commonly in the literatures, for example [8]–[14]. In addition, and Assumption 1 becomes the one adopted in [5], [15], and [16].

Let $T > 0$ be an arbitrary time. Denote by $t_1, \dots, t_{N_\sigma(T,0)}$ the switching times on the interval $(0, T)$ ($t_0 = 0$). When $t \in [t_l, t_{l+1})$, $\sigma(t) = k_l$, means that, the k_l th large-scale system is active. In this paper, we assume that $k_l \neq k_{l+1}$ for all l .

Lemma 1 [19], [20]: A switched nonlinear large-scale system (1) is called to have a switching signal $\sigma(t)$ with average dwell time τ_a if there exist two positive numbers N_0 and τ_a such that

$$N_\sigma(T, t) \leq N_0 + \frac{T - t}{\tau_a}, \quad \forall T \geq t \geq 0 \quad (8)$$

where $N_\sigma(T, t)$ is the number of switches occurring in the interval $[t, T)$.

B. Fuzzy Logic Systems

An FLS consists of four parts: a knowledge base, a fuzzifier, a fuzzy inference engine with fuzzy rules, and a defuzzifier. The knowledge base for FLS is comprised of a collection of fuzzy IF-THEN rules as follows:

R^l : If x_1 is F_1^l and x_2 is F_2^l and ... and x_n is F_n^l , then y is G^l , $l = 1, 2, \dots, N$

where $x = [x_1, \dots, x_n]^T$ and y are the FLS input and output, respectively. F_i^l and G^l are fuzzy sets with membership functions are $\mu_{F_i^l}(x_i)$ and $\mu_{G^l}(y)$, respectively. N is the number of rules.

Through singleton fuzzifier, center average defuzzification and product inference [30], the FLS can be expressed as

$$y(x) = \frac{\sum_{l=1}^N \bar{y}_l \prod_{i=1}^n \mu_{F_i^l}(x_i)}{\sum_{l=1}^N \left[\prod_{i=1}^n \mu_{F_i^l}(x_i) \right]}$$

where $\bar{y}_l = \max_{y \in R} \mu_{G^l}(y)$.

Define the fuzzy basis functions as

$$\varphi_l = \frac{\prod_{i=1}^n \mu_{F_i^l}(x_i)}{\sum_{l=1}^N \left(\prod_{i=1}^n \mu_{F_i^l}(x_i) \right)}.$$

Denoting $\theta^T = [\bar{y}_1, \bar{y}_2, \dots, \bar{y}_N] = [\theta_1, \theta_2, \dots, \theta_N]$ and $\varphi(x) = [\varphi_1(x), \dots, \varphi_N(x)]^T$, then FLS (6) can be rewritten as

$$y(x) = \theta^T \varphi(x). \quad (9)$$

Lemma 2 [32]: Let $f(x)$ be a continuous function defined on a compact set Ω . Then for any constant $\varepsilon > 0$, there exists an FLS $\hat{f}(x|\theta) = \theta^T \varphi(x)$ such as

$$\sup_{x \in \Omega} |f(x) - (\theta^*)^T \varphi(x)| \leq \varepsilon. \quad (10)$$

The control objective of this paper is to design an adaptive fuzzy decentralized output feedback control scheme such that all the variables in the switched closed-loop system are bounded and the system output y_i can track the given

reference signal $y_{i,r}(t)$ [$y_{i,r}(t)$ and its time derivatives up to the n_i th order are continuous and bounded] in presence of the unknown function $f_{i,j}^{\sigma(t)}(x_{i,j})$ and unmeasurable states $x_{i,j}$, $i = 1, 2, \dots, N$, $j = 2, \dots, n_i$.

III. SWITCHED FUZZY DECENTRALIZED STATE OBSERVER DESIGN

Since the states $x_{i,2}, x_{i,3}, \dots, x_{i,n_i-1}$ and x_{n_i} in the switched nonlinear large-scale system (1) are not available for feedback, this section aims to design a switched decentralized state observer for estimating the unmeasurable states in switched system (1).

Note that the functions $f_{i,j}^{\sigma(t)}(x_{i,j})$ are unknown. Thus the FLSs are utilized to approximate them. By Lemma 2, we assume that

$$\begin{aligned} f_{i,j}^{\sigma(t)}(x_{i,j}|\theta_{i,j}^{\sigma(t)}) &= \theta_{i,j}^{\sigma(t)T} \varphi_{i,j}^{\sigma(t)}(x_{i,j}) \\ \hat{f}_{i,j}^{\sigma(t)}(\hat{x}_{i,j}|\theta_{i,j}^{\sigma(t)}) &= \theta_{i,j}^{\sigma(t)T} \varphi_{i,j}^{\sigma(t)}(\hat{x}_{i,j}) \\ i &= 1, 2, \dots, N, \quad j = 1, 2, \dots, n_i \end{aligned} \quad (11)$$

where $\hat{x}_{i,j} = [\hat{x}_{i,1}, \hat{x}_{i,2}, \dots, \hat{x}_{i,n_i}]^T$ are the estimations of $x_{i,j} = [x_{i,1}, x_{i,2}, \dots, x_{i,n_i}]^T$, $i = 1, 2, \dots, N$.

According to definition of the optimal parameter vectors $\theta_{i,j}^{*\sigma(t)}$ in [32]–[34], let

$$\begin{aligned} \theta_{i,j}^{*\sigma(t)} &= \arg \min_{\theta_{i,j}^{\sigma(t)} \in \Omega_{i,j}^{\sigma(t)}} \left[\sup_{(x_{i,j} \times \hat{x}_{i,j}) \in U_{i,j,1} \times U_{i,j,2}} \right. \\ &\quad \left. \times \left| \hat{f}_{i,j}^{\sigma(t)}(\hat{x}_{i,j}|\theta_{i,j}^{\sigma(t)}) - f_{i,j}^{\sigma(t)}(x_{i,j}) \right| \right] \\ i &= 1, 2, \dots, N, \quad j = 1, 2, \dots, n_i \end{aligned} \quad (12)$$

where $\Omega_{i,j}^{\sigma(t)}$, $U_{i,j,1}$, and $U_{i,j,2}$ are bounded compact regions for $\theta_{i,j}^{\sigma(t)}$, $x_{i,j}$, and $\hat{x}_{i,j}$, respectively. In addition, the approximation error $\varepsilon_{i,j}^{\sigma(t)}$ is defined as

$$\begin{aligned} f_{i,j}^{\sigma(t)}(x_{i,j}) &= \hat{f}_{i,j}^{\sigma(t)}(\hat{x}_{i,j}|\theta_{i,j}^{*\sigma(t)}) + \varepsilon_{i,j}^{\sigma(t)} \\ i &= 1, 2, \dots, N, \quad j = 1, 2, \dots, n_i \end{aligned} \quad (13)$$

where $\varepsilon_{i,j}^{\sigma(t)}$ satisfies $|\varepsilon_{i,j}^{\sigma(t)}| \leq \varepsilon_{i,j}^{*\sigma(t)}$, with $\varepsilon_{i,j}^{*\sigma(t)}$ being an unknown constant.

By substituting (13) into (1), the switched system (1) can be expressed as

$$\begin{aligned} \dot{x}_{i,j} &= x_{i,j+1} + \hat{f}_{i,j}^{\sigma(t)}(\hat{x}_{i,j}|\theta_{i,j}^{*\sigma(t)}) + \varepsilon_{i,j}^{\sigma(t)} + \Delta_{i,j}^{\sigma(t)}(t) \\ i &= 1, 2, \dots, N, \quad j = 1, 2, \dots, n_i - 1 \\ \dot{x}_{i,n_i} &= D_i^{\sigma(t)}(u_i^{\sigma(t)}) + \hat{f}_{i,n_i}^{\sigma(t)}(\hat{x}_{i,n_i}|\theta_{i,j}^{*\sigma(t)}) + \varepsilon_{i,n_i}^{\sigma(t)} + \Delta_{i,n_i}^{\sigma(t)}(t) \\ y_i &= x_{i,1}. \end{aligned} \quad (14)$$

Rewrite (14) as

$$\begin{aligned} \dot{\underline{x}}_{i,n_i} &= A_i^{\sigma(t)} \underline{x}_{i,n_i} + L_i^{\sigma(t)} y_i + \sum_{j=1}^{n_i} B_{i,j} \theta_{i,j}^{*\sigma(t)T} \varphi_{i,j}^{\sigma(t)}(\hat{x}_{i,j}) \\ &\quad + \varepsilon_i^{\sigma(t)} + \Delta_i^{\sigma(t)} + B_{i,n_i} D_i^{\sigma(t)}(u_i^{\sigma(t)}) \\ y_i &= C_i \underline{x}_{i,n_i} \end{aligned} \quad (15)$$

where

$$\begin{aligned} A_i^{\sigma(t)} &= \begin{bmatrix} -l_{i,1}^{\sigma(t)} \\ \vdots \\ -l_{i,n_i}^{\sigma(t)} & 0 & \dots & 0 \end{bmatrix}, \quad L_i^{\sigma(t)} = \begin{bmatrix} l_{i,1}^{\sigma(t)} \\ \vdots \\ l_{i,n_i}^{\sigma(t)} \end{bmatrix} \\ \Delta_i^{\sigma(t)} &= \begin{bmatrix} \Delta_{i,1}^{\sigma(t)}(\bar{y}) \\ \vdots \\ \Delta_{i,n_i}^{\sigma(t)}(\bar{y}) \end{bmatrix}, \quad C_i = [1, \dots, 0] \\ \varepsilon_i^{\sigma(t)} &= [\varepsilon_{i,1}^{\sigma(t)}, \varepsilon_{i,2}^{\sigma(t)}, \dots, \varepsilon_{i,n_i}^{\sigma(t)}]^T \\ B_{i,j} &= \begin{bmatrix} 0, \dots, 0, 1, 0, \dots, 0 \end{bmatrix}_j^T. \end{aligned}$$

To estimate the states of the system, the following switched fuzzy decentralized state observer is designed:

$$\begin{aligned} \dot{\hat{x}}_{i,n_i} &= A_i^{\sigma(t)} \hat{x}_{i,n_i} + L_i^{\sigma(t)} y_i + \sum_{j=1}^{n_i} B_{i,j} \hat{f}_{i,j}^{\sigma(t)}(\hat{x}_{i,j}|\theta_{i,j}^{\sigma(t)}) \\ &\quad + B_{i,n_i} D_i^{\sigma(t)}(u_i^{\sigma(t)}) \\ \hat{y}_i &= C_i \hat{x}_{i,n_i}. \end{aligned} \quad (16)$$

Define observation error vector e_i as

$$e_i = [e_{i,1}, e_{i,2}, \dots, e_{i,n_i}]^T = \underline{x}_{i,n_i} - \hat{x}_{i,n_i}. \quad (17)$$

From (15)–(17), the observation error can be expressed as

$$\dot{e}_i = A_i^{\sigma(t)} e_i + \sum_{j=1}^{n_i} B_{i,j} (\tilde{\theta}_{i,j}^{\sigma(t)})^T \varphi_{i,j}^{\sigma(t)}(\hat{x}_{i,j}) + \varepsilon_i^{\sigma(t)} + \Delta_i^{\sigma(t)} \quad (18)$$

where $\tilde{\theta}_{i,j}^{\sigma(t)} = \theta_{i,j}^{*\sigma(t)} - \theta_{i,j}^{\sigma(t)}$, $i = 1, 2, \dots, N$, and $j = 1, 2, \dots, n_i$ are the adaptive parameter vector errors.

For the switching signal $\sigma(t) = k$, $k = 1, 2, \dots, m$; the k_j th switched large-scale system is active and the remaining large-scale systems are inactive. Choose vector L_i^k such that matrix A_i^k is a strict Hurwitz matrix, therefore, for any a given matrix $Q_i^k = (Q_i^k)^T > 0$, there exists a positive definite matrix $P_i^k = (P_i^k)^T > 0$ such that

$$(A_i^k)^T P_i^k + P_i^k A_i^k = -Q_i^k. \quad (19)$$

IV. ADAPTIVE FUZZY CONTROL DESIGN

Based on the fuzzy decentralized state observer designed in Section III, an adaptive fuzzy decentralized output feedback control is proposed in this section via backstepping technique. The stability of the whole closed-loop system will be proved by using Lyapunov function and average dwell-time method.

When $t \in [t_j, t_{j+1})$, and $\sigma(t) = k$, the k th switched large-scale system is active. For the k th activated large-scale system, the n_i -step adaptive fuzzy decentralized backstepping output feedback control design is based on the following change of coordinates:

$$\begin{aligned} z_{i,1} &= y_i - y_{i,r} \\ z_{i,j} &= \hat{x}_{i,j} - \alpha_{i,j-1}^k, \quad i = 1, 2, \dots, N, \quad j = 2, \dots, n_i \end{aligned} \quad (20)$$

where $\alpha_{i,j-1}^k$ is the intermediate control function.

Step i, 1: Given $x_{i,2}$ in terms of its estimation as $\hat{x}_{i,2} = \hat{x}_{i,2} + e_{i,2}$, we obtain

$$\begin{aligned}\dot{z}_{i,1} &= x_{i,2} + f_{i,1}^k(\underline{x}_{i,1}) + \Delta_{i,1}^k(\bar{y}) - \dot{y}_{i,r} \\ &= z_{i,2} + \alpha_{i,1}^k + e_{i,2} + \varepsilon_{i,1}^k + \left(\theta_{i,1}^k\right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) \\ &\quad + \left(\tilde{\theta}_{i,1}^k\right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) + \Delta_{i,1}^k(\bar{y}) - \dot{y}_{i,r}. \quad (21)\end{aligned}$$

Choose the intermediate control function $\alpha_{i,1}^k$, and the adaptation function $\theta_{i,1}^k$ as follows:

$$\alpha_{i,1}^k = -c_{i,1}^k z_{i,1} - \Psi_{i,1}^k z_{i,1} - \left(\theta_{i,1}^k\right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) - \dot{y}_{i,r} \quad (22)$$

$$\dot{\theta}_{i,1}^k = \delta_{i,1}^k \varphi_{i,1}^k(\hat{x}_{i,1}) z_{i,1} - \tau_{i,1}^k \theta_{i,1}^k \quad (23)$$

where $c_{i,1}^k > 0$, $\delta_{i,1}^k > 0$, and $\tau_{i,1}^k > 0$ are the design parameters, $\theta_{i,1}^k$ is the estimation of $\theta_{i,1}^{*k}$, $\Psi_{i,1}^k = 2 + 2n_i \sum_{l=1}^N \bar{h}_{l,i}^k$.

Step i, j, (2 ≤ j ≤ n_i): Since $z_{i,j} = \hat{x}_{i,j} - \alpha_{i,j-1}^k$, and similar to the procedures in the first step, the time derivative of $z_{i,j}$ along with (16) and (20) is

$$\begin{aligned}\dot{z}_{i,j} &= z_{i,j+1} + \alpha_{i,j}^k + l_{i,j}^k e_{i,1} + \left(\theta_{i,j}^k\right)^T \varphi_{i,j}^k(\hat{x}_{i,j}) \\ &\quad - \sum_{p=1}^{j-1} \frac{\partial \alpha_{i,j-1}^k}{\partial \hat{x}_{i,p}^k} \dot{\hat{x}}_{i,p}^k - \sum_{p=1}^{j-1} \frac{\partial \alpha_{i,j-1}^k}{\partial \theta_{i,j}^k} \dot{\theta}_{i,j}^k - \sum_{p=1}^{j-1} \frac{\partial \alpha_{i,j-1}^k}{\partial y_{i,r}^{(p)}} y_{i,r}^{(p)} \\ &\quad - \frac{\partial \alpha_{i,j-1}^k}{\partial x_{i,1}} \left[\hat{x}_{i,2} + e_{i,2} + \left(\theta_{i,j}^k\right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) \right. \\ &\quad \left. + \left(\tilde{\theta}_{i,j}^k\right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) + \varepsilon_{i,1}^k + \Delta_{i,1}^k(\bar{y}) \right] \\ &= z_{i,j+1} + \alpha_{i,j}^k + \Xi_{i,j}^k \\ &\quad - \frac{\partial \alpha_{i,j-1}^k}{\partial x_{i,1}} \left(\left(\tilde{\theta}_{i,j}^k\right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) + \varepsilon_{i,1}^k + e_{i,2} + \Delta_{i,1}^k(\bar{y}) \right) \quad (24)\end{aligned}$$

where

$$\begin{aligned}\Xi_{i,j}^k &= l_{i,j}^k e_{i,1} + \left(\theta_{i,j}^k\right)^T \varphi_{i,j}^k(\hat{x}_{i,j}) - \sum_{p=1}^{j-1} \frac{\partial \alpha_{i,j-1}^k}{\partial \hat{x}_{i,p}^k} \dot{\hat{x}}_{i,p}^k \\ &\quad - \sum_{p=1}^{j-1} \frac{\partial \alpha_{i,j-1}^k}{\partial \theta_{i,p}^k} \dot{\theta}_{i,p}^k - \sum_{p=1}^{j-1} \frac{\partial \alpha_{i,j-1}^k}{\partial y_{i,r}^{(p)}} y_{i,r}^{(p)} \\ &\quad - \frac{\partial \alpha_{i,j-1}^k}{\partial x_{i,1}} \left[\hat{x}_{i,2} + \left(\theta_{i,j}^k\right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) \right].\end{aligned}$$

Choose the intermediate control function $\alpha_{i,j}^k$, and the adaptation function $\theta_{i,j}^k$ as follows:

$$\alpha_{i,j}^k = -c_{i,j}^k z_{i,j} - \frac{3}{2} z_{i,j} - 2 \left(\frac{\partial \alpha_{i,j-1}^k}{\partial x_{i,1}} \right)^2 z_{i,j} - \Xi_{i,j}^k \quad (25)$$

$$\dot{\theta}_{i,j}^k = \delta_{i,j}^k z_{i,j} \varphi_{i,j}^k(\hat{x}_{i,j}) - \tau_{i,j}^k \theta_{i,j}^k \quad (26)$$

where $c_{i,j}^k > 0$, $\delta_{i,j}^k > 0$, and $\tau_{i,j}^k > 0$ are the design parameters, $\theta_{i,j}^k$ is the estimation of $\theta_{i,j}^{*k}$.

Step i, n_i: In the final design step n_i , the actual decentralized control input u_i^k appears. The time derivative of z_{i,n_i} is

$$\begin{aligned}\dot{z}_{i,n_i} &= \dot{\hat{x}}_{i,n_i} - \dot{\alpha}_{i,n_i-1}^k = D_i^k(u_i^k) + \Xi_{i,n_i}^k \\ &\quad - \frac{\partial \alpha_{i,n_i-1}^k}{\partial x_{i,1}} \left(\left(\tilde{\theta}_{i,1}^k\right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) + \varepsilon_{i,1}^k + e_{i,2} + \Delta_{i,1}^k(\bar{y}) \right) \\ &= u_{i,d}^k + \left(\tilde{d}_{i,r,m}^k - \frac{u_{i,d}^k + \hat{d}_{i,r,m}^k \tilde{m}_{i,r}^k}{\hat{m}_{i,r}^k} \right) \delta_i^k \\ &\quad + \left(\tilde{d}_{i,l,m}^k - \frac{u_{i,d}^k + \hat{d}_{i,l,m}^k \tilde{m}_{i,l}^k}{\hat{m}_{i,l}^k} \right) (1 - \delta_i^k) + \Xi_{i,n_i}^k \\ &\quad - \frac{\partial \alpha_{i,n_i-1}^k}{\partial x_{i,1}} \left(\left(\tilde{\theta}_{i,1}^k\right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) \right. \\ &\quad \left. + \varepsilon_{i,1}^k + e_{i,2} + \Delta_{i,1}^k(\bar{y}) \right) + \varepsilon_{i,d}^k \quad (27)\end{aligned}$$

where

$$\begin{aligned}\Xi_{i,n_i}^k &= l_{i,n_i}^k e_{i,1} + \left(\theta_{i,n_i}^k\right)^T \varphi_{i,n_i}^k(\hat{x}_{i,n_i}) - \sum_{j=1}^{n_i-1} \frac{\partial \alpha_{i,n_i-1}^k}{\partial \hat{x}_{i,j}^k} \dot{\hat{x}}_{i,j}^k \\ &\quad - \sum_{j=1}^{n_i-1} \frac{\partial \alpha_{i,n_i-1}^k}{\partial \theta_{i,j}^k} \dot{\theta}_{i,j}^k - \sum_{j=1}^{n_i-1} \frac{\partial \alpha_{i,n_i-1}^k}{\partial y_{i,r}^{(j)}} y_{i,r}^{(j)} \\ &\quad - \frac{\partial \alpha_{i,n_i-1}^k}{\partial x_{i,1}} \left[\hat{x}_{i,2} + \left(\theta_{i,1}^k\right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) \right].\end{aligned}$$

The decentralized controller $u_{i,d}^k$, and adaptation function θ_{i,n_i}^k , $\hat{m}_{i,l}^k$, $\hat{m}_{i,r}^k$, $\hat{d}_{i,l,m}^k$, and $\hat{d}_{i,r,m}^k$ are chosen as

$$u_i^k = -c_{i,n_i}^k z_{i,n_i} - z_{i,n_i} - 2 \left(\frac{\partial \alpha_{i,n_i-1}^k}{\partial x_{i,1}} \right)^2 z_{i,n_i} - \Xi_{i,n_i}^k \quad (28)$$

$$\dot{\theta}_{i,n_i}^k = \delta_{i,n_i}^k z_{i,n_i} \varphi_{i,n_i}^k(\hat{x}_{i,n_i}) - \tau_{i,n_i}^k \theta_{i,n_i}^k \quad (29)$$

$$\hat{m}_{i,l}^k = \frac{u_{i,d}^k + \hat{d}_{i,l,m}^k}{\hat{m}_{i,l}^k} z_{i,n_i} \eta_{i,1}^k (1 - \delta_i^k) + \tau_{i,n_i}^k \hat{m}_{i,l}^k \quad (30)$$

$$\hat{m}_{i,r}^k = \frac{u_{i,d}^k + \hat{d}_{i,r,m}^k}{\hat{m}_{i,r}^k} z_{i,n_i} \eta_{i,2}^k \delta_i^k + \tau_{i,n_i}^k \hat{m}_{i,r}^k \quad (31)$$

$$\hat{d}_{i,l,m}^k = -z_{i,n_i} \eta_{i,3}^k (1 - \delta_i^k) + \tau_{i,n_i}^k \hat{d}_{i,l,m}^k \quad (32)$$

$$\hat{d}_{i,r,m}^k = z_{i,n_i} \eta_{i,4}^k \delta_i^k + \tau_{i,n_i}^k \hat{d}_{i,r,m}^k \quad (33)$$

where $c_{i,n_i}^k > 0$, $\delta_{i,n_i}^k > 0$, $\tau_{i,n_i}^k > 0$, and $\eta_{i,a}^k > 0$ ($a = 1, 2, 3, 4$) are design parameters. θ_{i,n_i}^k , $\hat{m}_{i,r}^k$, $\hat{m}_{i,l}^k$, $\hat{d}_{i,r,m}^k$, and $\hat{d}_{i,l,m}^k$ are the estimation of θ_{i,n_i}^{*k} , $m_{i,r}^k$, $m_{i,l}^k$, $m_{i,r}^k d_{i,r}^k$, and $m_{i,l}^k d_{i,l}^k$.

V. STABILITY ANALYSIS

Theorem 1: For switched uncertain nonlinear large-scale system (1), under Assumption 1, if the average dwell time of the switching signal satisfies $\tau_a > \ln \mu / C$, then the decentralized controller (8), switched fuzzy decentralized state observer (16), together with the intermediate control functions (22) and (25), and parameter adaptation functions (23), (26), and (29) can guarantee that all the signals in the closed-loop system are bounded. Moreover, the observation errors and the tracking errors can be made to converge to a small neighborhood of zero with a proper selection design parameters.

Proof: From (20), (21), (24), and (27), we get

$$z_{i,1}\dot{z}_{i,1} = z_{i,1} \left(z_{i,2} + \alpha_{i,1}^k + \varepsilon_{i,1}^k + \left(\tilde{\theta}_{i,1}^k \right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) + e_{i,2} + \left(\tilde{\theta}_{i,1}^k \right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) + \Delta_{i,1}^k(\bar{y}) - \dot{y}_{i,r} \right) \quad (34)$$

$$z_{i,j}\dot{z}_{i,j} = z_{i,j} \left(z_{i,j+1} + \alpha_{i,j}^k + \Xi_{i,j}^k - \frac{\partial \alpha_{i,j-1}^k}{\partial x_{i,1}} \left(\varepsilon_{i,1}^k + e_{i,2} + \left(\tilde{\theta}_{i,1}^k \right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) + \Delta_{i,1}^k(\bar{y}) \right) \right) \quad (35)$$

$$z_{i,n_i}\dot{z}_{i,n_i} = z_{i,n_i} \left(u_{i,d}^k + \left(\tilde{d}_{i,r,m}^k - \frac{u_{i,d}^k + \hat{d}_{i,r,m}^k}{\hat{m}_{i,r}^k} \tilde{m}_{i,r}^k \right) \delta_i^k + \left(\tilde{d}_{i,l,m}^k - \frac{u_{i,d}^k + \hat{d}_{i,l,m}^k}{\hat{m}_{i,l}^k} \tilde{m}_{i,l}^k \right) (1 - \delta_i^k) + \Xi_{i,n_i}^k - \frac{\partial \alpha_{i,n_i-1}^k}{\partial x_{i,1}} \left(\left(\tilde{\theta}_{i,1}^k \right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) + \varepsilon_{i,1}^k + e_{i,2} + \Delta_{i,1}^k(\bar{y}) \right) + \varepsilon_{i,d}^k \right). \quad (36)$$

By Assumption 1, Young's inequality and $y_i = z_{i,1} + y_{i,r}$, we can obtain

$$\begin{aligned} \sum_{i=1}^N 2e_i^T P_i^k \Delta_i^k &\leq \sum_{i=1}^N \left(\|e_i\|^2 \|P_i^k\|^2 + \|\Delta_i^k\|^2 \right) \\ &\leq \sum_{i=1}^N \|e_i\|^2 \|P_i^k\|^2 \\ &\quad + n_i \sum_{l=1}^N \sum_{i=1}^N h_{l,i}^k (y_{i,r}^2 + z_{i,1}^2) \\ &\quad + 2n_i \left(\sum_{i=1}^N \sum_{l=1}^N \beta_l^k(0) \right)^2 \end{aligned} \quad (37)$$

$$\sum_{j=1}^{n_i} B_{i,j} \left(\tilde{\theta}_{i,1}^k \right)^T \varphi_{i,j}^k(\hat{x}_{i,j}) \leq n_i \|P_i^k\|^2 \|e_i\|^2 + \sum_{j=1}^{n_i} \left(\tilde{\theta}_{i,j}^k \right)^T \tilde{\theta}_{i,j}^k \quad (38)$$

$$\begin{aligned} \sum_{i=1}^N z_{i,1} \Delta_{i,1}^k(\bar{y}) &\leq \sum_{i=1}^N \frac{1}{2} z_{i,1}^2 + \sum_{i=1}^N \sum_{l=1}^N h_{l,i}^k (y_{i,r}^2 + z_{i,1}^2) \\ &\quad + \left(\sum_{i=1}^N \sum_{l=1}^N \beta_l^k(0) \right)^2 \end{aligned} \quad (39)$$

$$\begin{aligned} - \sum_{i=1}^N z_{i,j} \frac{\partial \alpha_{i,j-1}^k}{\partial x_{i,1}} \Delta_{i,1}^k(\bar{y}) &\leq \sum_{i=1}^N \frac{1}{2} z_{i,j}^2 \left(\frac{\partial \alpha_{i,j-1}^k}{\partial x_{i,1}} \right)^2 \\ &\quad + \sum_{i=1}^N \sum_{l=1}^N h_{l,i}^k (y_{i,r}^2 + z_{i,1}^2) \\ &\quad + \left(\sum_{i=1}^N \sum_{l=1}^N \beta_l^k(0) \right)^2. \end{aligned} \quad (40)$$

By Young's inequality and the fact $\varphi_{i,j}^k(\hat{x}_{i,j})^T \varphi_{i,j}^k(\hat{x}_{i,j}) \leq 1$, we can obtain

$$2e_i^T P_i^k \varepsilon_i^k \leq \|P_i^k\| \|e_i\|^2 + \|\varepsilon_i^k\|^2 \quad (41)$$

$$z_{i,1} \left(z_{i,2} + \varepsilon_{i,2}^{*k} + e_{i,2} \right) \leq \frac{3}{2} z_{i,1}^2 + \frac{1}{2} z_{i,2}^2 + \|e_i\|^2 + \frac{1}{2} \|\varepsilon_i^{*k}\|^2 \quad (42)$$

$$\begin{aligned} -z_{i,j} \frac{\partial \alpha_{i,j-1}^k}{\partial x_{i,1}} \left(\varepsilon_{i,1}^k + e_{i,2} \right) &\leq \frac{1}{2} \left(\varepsilon_{i,1}^k \right)^2 + \frac{1}{2} \|e_i\|^2 \\ &\quad + \left(\frac{\partial \alpha_{i,j-1}^k}{\partial x_{i,1}} \right)^2 z_{i,j}^2 \end{aligned} \quad (43)$$

$$\begin{aligned} -z_{i,j} \frac{\partial \alpha_{i,j-1}^k}{\partial x_{i,1}} \left(\tilde{\theta}_{i,j}^k \right)^T \varphi_{i,1}^k(\hat{x}_{i,1}) &\leq \frac{1}{2} \left(\tilde{\theta}_{i,j}^k \right)^T \tilde{\theta}_{i,1}^k \\ &\quad + \frac{1}{2} \left(\frac{\partial \alpha_{i,j-1}^k}{\partial x_{i,1}} \right)^2 z_{i,j}^2 \end{aligned} \quad (44)$$

$$\begin{aligned} z_{i,j} \left(z_{i,j+1} - \left(\tilde{\theta}_{i,j}^k \right)^T \varphi_{i,j}^k(\hat{x}_{i,j}) \right) &\leq z_{i,j}^2 + \frac{1}{2} z_{i,j+1}^2 \\ &\quad + \frac{1}{2} \left(\tilde{\theta}_{i,j}^k \right)^T \tilde{\theta}_{i,j}^k \end{aligned} \quad (45)$$

where $\varepsilon_i^{*k} = [\varepsilon_{i,1}^{*k}, \varepsilon_{i,2}^{*k}, \dots, \varepsilon_{i,n_i}^{*k}]^T$.

Consider the total Lyapunov candidate V^k as the sum of local Lyapunov candidate functions

$$V^k = \sum_{i=1}^N \sum_{j=1}^{n_i} V_{i,j}^k \quad (46)$$

where $V_{i,1}^k = e_i^T P_i^k e_i + 1/2 z_{i,1}^2 + 1/2 \delta_{i,1}^k (\tilde{\theta}_{i,1}^k)^T \tilde{\theta}_{i,1}^k$, $V_{i,j}^k = 1/2 z_{i,j}^2 + 1/2 \delta_{i,j}^k (\tilde{\theta}_{i,j}^k)^T \tilde{\theta}_{i,j}^k$, and $V_{i,n_i}^k = 1/2 z_{i,n_i}^2 + 1/2 \delta_{i,n_i}^k (\tilde{\theta}_{i,n_i}^k)^T \tilde{\theta}_{i,n_i}^k + 1/2 \eta_{i,1}^k (\tilde{m}_{i,l}^k)^2 + 1/2 \eta_{i,2}^k (\tilde{m}_{i,r}^k)^2 + 1/2 \eta_{i,3}^k (\tilde{d}_{i,l,m}^k)^2 + 1/2 \eta_{i,4}^k (\tilde{d}_{i,r,m}^k)^2$, $i = 1, \dots, N, j = 2, \dots, n_i - 1$.

From (22), (23), (25), (26), (28)–(33), and (37)–(45), we can obtain

$$\begin{aligned} \dot{V}_{i,n_i}^k &\leq -\lambda_{i,n_i}^k \|e_i\|^2 - \sum_{j=1}^{n_i} c_{i,j}^k z_{i,j}^2 + \sum_{j=1}^{n_i} \frac{\tau_{i,j}^k}{\delta_{i,j}^k} \left(\tilde{\theta}_{i,j}^k \right)^T \tilde{\theta}_{i,j}^k \\ &\quad + 2n_i \sum_{l=1}^N h_{l,i}^k y_{i,r}^2 + \sum_{j=1}^{n_i} \left(\tilde{\theta}_{i,j}^k \right)^T \tilde{\theta}_{i,j}^k + \frac{1}{2} \sum_{j=2}^{n_i} \left(\tilde{\theta}_{i,j}^k \right)^T \tilde{\theta}_{i,j}^k \\ &\quad + \frac{n_i - 1}{2} \left(\tilde{\theta}_{i,1}^k \right)^T \tilde{\theta}_{i,1}^k + \frac{\tau_{i,n_i}^k}{\eta_{i,k}} \tilde{m}_{i,l}^k \hat{m}_{i,l}^k + \frac{\tau_{i,n_i}^k}{\eta_{i,2}^k} \tilde{m}_{i,r}^k \hat{m}_{i,r}^k \\ &\quad + \frac{\tau_{i,n_i}^k}{\eta_{i,3}^k} \tilde{d}_{i,l,m}^k \hat{d}_{i,l,m}^k + \frac{\tau_{i,n_i}^k}{\eta_{i,4}^k} \tilde{d}_{i,r,m}^k \hat{d}_{i,r,m}^k + D_{i,n_i}^k \end{aligned} \quad (47)$$

where $\lambda_{i,n_i}^k = \lambda_{\min}(Q_i) - 2\|P_i^k\|^2 - n_i\|P_i^k\|^2 - 1 - n_i/2$, and $D_{i,n_i}^k = n_i - 1/2\|\varepsilon_{i,j}^{*k}\|^2 + 3n_i(\sum_{l=1}^N \beta_l(0))^2 + (\bar{\varepsilon}_{i,d}^k)^2$, $|\varepsilon_{i,d}^k| \leq \bar{\varepsilon}_{i,d}^k$, and $\bar{\varepsilon}_{i,d}^k$ is a constant.

By completing the square for each parameter estimation, (47) can be rewritten as

$$\begin{aligned} \dot{V}^k \leq & \sum_{i=1}^N \left\{ -\lambda_{i,n_i}^k \|e_i\|^2 - \sum_{j=1}^{n_i} c_{i,j}^k z_{i,j}^2 \right. \\ & - \frac{1}{2} \sum_{j=1}^{n_i} \left(\frac{\tau_{i,j}^k}{\delta_{i,j}^k} - 3 \right) \left(\tilde{\theta}_{i,j}^k \right)^T \tilde{\theta}_{i,j}^k - \frac{\tau_{i,n_i}^k}{2\eta_{1,k}} \left(\tilde{m}_{i,l}^k \right)^2 \\ & - \frac{1}{2} \left(\frac{\tau_{i,1}^k}{\delta_{i,1}^k} - n_i - 1 \right) \left(\tilde{\theta}_{i,1}^k \right)^T \tilde{\theta}_{i,1}^k - \frac{\tau_{i,n_i}^k}{2\eta_{i,2}^k} \left(\tilde{m}_{i,r}^k \right)^2 \\ & \left. - \frac{\tau_{i,n_i}^k}{2\eta_{i,3}^k} \left(\tilde{d}_{i,l,m}^k \right)^2 - \frac{\tau_{i,n_i}^k}{2\eta_{i,4}^k} \left(\tilde{d}_{i,r,m}^k \right)^2 + D_{i,n_i}^k \right\} \quad (48) \end{aligned}$$

where $D_i^k = D_{i,n_i-1}^k + \sum_{j=1}^{n_i} (\tau_{i,j}^k / 2\delta_{i,j}^k) (\theta_{i,j}^k)^* (\theta_{i,j}^k)^* + 2n_i \sum_{l=1}^N h_{i,l}^k y_{i,M}^2 + (\tau_{i,n_i}^k / 2\eta_{1,k}) (m_{i,l}^k)^2 + (\tau_{i,n_i}^k / 2\eta_{i,2}^k) (m_{i,r}^k)^2 + (\tau_{i,n_i}^k / 2\eta_{i,3}^k) (d_{i,l,m}^k)^2 + (\tau_{i,n_i}^k / 2\eta_{i,4}^k) (d_{i,r,m}^k)^2$ and $y_{i,r}^2 \leq y_{i,M}^2$.

Let $C_i^k = \min\{\lambda_{i,n_i}^k / \lambda_{i,\max}(P_i^k), c_{i,j}^k, \tau_{i,j}^k - 3\delta_{i,j}^k, \tau_{i,1}^k - \delta_{i,1}^k n_i - \delta_{i,1}^k, \tau_{i,n_i}^k\}$, $(\tau_{i,j}^k / \delta_{i,j}^k) - 3 > 0$, $c_{i,j}^k > 0$, $\tau_{i,1}^k / \delta_{i,1}^k - n_i - 1 > 0$.

Then (41) can be rewritten as

$$\dot{V}^k \leq -CV^k + D \quad (49)$$

where $C = \min_{k \in M} \sum_{i=1}^N C_i^k$, $D = \max_{k \in M} \sum_{i=1}^N D_i^k$. ■

Remark 2: It is worth pointing out that the switched systems are different from those nonswitched systems, which brings more difficulties in the control design of switched systems than that of nonswitched systems. The main problems are: 1) how to design switched laws to ensure a stable closed loop control system and 2) how to prove the stability of the closed-loop systems.

It is easy to see that the function $W(t) = e^{Ct} V^{\sigma(t)}(x(t))$ is piecewise differentiable along solutions of the system (1). In view of (49), on each interval $[t_j, t_{j+1})$, one has

$$\begin{aligned} \dot{W}(t) &= Ce^{Ct} V_{\sigma(t)}(x(t)) + e^{Ct} \dot{V}^{\sigma(t)}(x(t)) \\ &\leq De^{Ct}, t \in [t_j, t_{j+1}). \end{aligned} \quad (50)$$

With the results in [17] and [35], we have $V^k(x(t)) \leq \mu V^l(x(t))$ ($\mu > 1, k, l \in M$), which implies that

$$\begin{aligned} W(t_{j+1}) &= e^{Ct_{j+1}} V^{\sigma(t_{j+1})}(x(t_{j+1})) \\ &\leq \mu e^{Ct_{j+1}} V^{\sigma(t_j)}(x(t_{j+1})) = \mu W(t_{j+1}^-) \\ &\leq \mu \left[W(t_j) + \int_{t_j}^{t_{j+1}} De^{Ct} dt \right]. \end{aligned} \quad (51)$$

For an arbitrary $T > t_0 = 0$, after iterating the inequality (51) from $j = 0$ to $j = N_{\sigma}(T, 0) - 1$, we obtain that

$$\begin{aligned} W(T^-) &\leq W(t_{N_{\sigma}(T, 0)}) + \int_{t_{N_{\sigma}(T, 0)}}^T De^{Ct} dt \\ &\leq \mu \left[W(t_{N_{\sigma}(T, 0)-1}) + \int_{t_{N_{\sigma}(T, 0)-1}}^{t_{N_{\sigma}(T, 0)}} De^{Ct} dt \right. \\ &\quad \left. + \mu^{-1} \int_{t_{N_{\sigma}(T, 0)}}^T De^{Ct} dt \right] \\ &\leq \dots \end{aligned}$$

$$\begin{aligned} &\leq \mu^{N_{\sigma}(T, 0)} \left[W(0) + \sum_{j=0}^{N_{\sigma}(T, 0)-1} \mu^{-j} \int_{t_j}^{t_{j+1}} De^{Ct} dt \right. \\ &\quad \left. + \mu^{-N_{\sigma}(T, 0)} \int_{t_{N_{\sigma}(T, 0)}}^T De^{Ct} dt \right]. \end{aligned} \quad (52)$$

Since $\tau_a > \ln \mu / C$, for any $\delta \in (0, C - \ln \mu / \tau_a)$, one has $\tau_a > \ln \mu / (C - \delta)$. By (8), it holds that

$$N_{\sigma}(T, t) \leq N_0 + \frac{(C - \delta)(T - t)}{\ln \mu}, \quad \forall T \geq t \geq 0. \quad (53)$$

In addition, it is clear that $N_{\sigma}(T, 0) - j \leq 1 + N_{\sigma}(T, t_{j+1})$, $j = 0, 1, \dots, N_{\sigma}(T, 0)$, one has

$$\mu^{N_{\sigma}(T, 0)-j} \leq \mu^{1+N_0} e^{(C-\delta)(T-t_{j+1})}. \quad (54)$$

In addition, since $\delta < C$ and

$$\int_{t_j}^{t_{j+1}} De^{Ct} dt \leq e^{(C-\delta)t_{j+1}} \int_{t_j}^{t_{j+1}} e^{(C-\delta)t} dt. \quad (55)$$

From (54) and (55), it then follows that:

$$W(T^-) \leq \mu^{N_{\sigma}(T, 0)} W(0) + \mu^{1+N_0} e^{(C-\delta)T} \int_0^T De^{\delta t} dt. \quad (56)$$

According to [26]–[29], there exist two class κ functions $\underline{\alpha}(|x|)$ and $\bar{\alpha}(|x|)$, which satisfy $\underline{\alpha}(|x|) \leq V^k(x) \leq \bar{\alpha}(|x|)$.

It indicates that

$$\begin{aligned} \underline{\alpha}(\|x(T)\|) &\leq V^{\sigma(T^-)}(x(T^-)) \\ &\leq e^{N_0 \ln \mu} e^{\left(\frac{\ln \mu}{\tau_a} - C\right)T} \bar{\alpha}(\|x(0)\|) \\ &\quad + \mu^{1+N_0} \frac{D}{\delta} (1 - e^{-\delta T}) \\ &\leq e^{N_0 \ln \mu} e^{\left(\frac{\ln \mu}{\tau_a} - C\right)T} \bar{\alpha}(\|x(0)\|) \\ &\quad + \mu^{1+N_0} \frac{D}{\delta}, \quad \forall T > 0. \end{aligned} \quad (57)$$

We conclude that, with (57) $\delta > 0$, and bounded initial conditions, if $\tau_a > (\ln \mu / C)$, then for bounded initial conditions, $e_{i,j}$, $z_{i,j}$, $\tilde{\theta}_{i,j}^k$ are bounded, since $\theta_{i,j}^{*k}$ are constants, $\theta_{i,j}^k$ are bounded. Further, it is easy to obtain that $\hat{x}_{i,j}$, $x_{i,j}$, and u_i^k , $i = 1, 2, \dots, N; j = 1, 2, \dots, n_i$, are also bounded. Hence, for bounded initial conditions, all the signals in the closed-loop system (1) are bounded for switching signal $\sigma(t)$ with average dwell time $\tau_a > (\ln \mu / C)$. From (57), it follows that

$$\begin{aligned} \frac{1}{2} \sum_{i=1}^N z_{i,1}^2(T) &\leq e^{N_0 \ln \mu} e^{\left(\frac{\ln \mu}{\tau_a} - C\right)T} \bar{\alpha}(\|x(0)\|) \\ &\quad + \mu^{1+N_0} \frac{D}{\delta} (1 - e^{-\delta T}), \quad \forall T > 0 \end{aligned}$$

which, together with $\tau_a > (\ln \mu / C)$, implies that

$$\begin{aligned} \lim_{t \rightarrow \infty} \sum_{i=1}^N z_{i,1}^2(t) &\leq e^{N_0 \ln \mu} e^{\left(\frac{\ln \mu}{\tau_a} - C\right)T} \bar{\alpha}(\|x(0)\|) \\ &\leq 2\mu^{1+N_0} \frac{D}{\delta} \leq \varsigma^2. \end{aligned}$$

This completes the proof of Theorem 1.

From previous discussions, the control design procedures and the guidelines for selecting the parameters are given as follows.

Step 1: Define the fuzzy IF-THEN rules and the membership functions, construct the fuzzy basis functions, and obtain the FLS (10).

Step 2: Specify the vector $L_i^k = [l_{i,1}^k, \dots, l_{i,n_i}^k]^T$, such that A_i^k are Hurwitz matrices.

Step 3: Specify a positive-definite matrices Q_i^k . By solving the Lyapunov equation (19), positive definite matrices P_i^k can be obtained.

Step 4: Select appropriately design parameters such that $c_{i,j}^k > 0$, $\delta_{i,j}^k > 0$, and $\tau_{i,j}^k > 0$, and the determine intermediate control functions $\alpha_{i,1}^k$ (22), $\alpha_{i,j}^k$ (25), and the parameters adaptation laws $\theta_{i,1}^k$ (23), and $\theta_{i,j}^k$ (26), $i = 1, \dots, N$; $j = 1, \dots, n_i - 1$; $k \in M$.

Step 5: Select appropriately design parameters $c_{i,n_i}^k > 0$ and $\delta_{i,n_i}^k > 0$, actual decentralized controller u_i^k (28), and adaptive update law θ_{i,n_i}^k (29), $\hat{m}_{i,l}^k$ (30), $\hat{m}_{i,r}^k$ (31), $\hat{d}_{i,l,m}^k$ (32), and $\hat{d}_{i,r,m}^k$ (33).

Remark 3: Note that from (57), we can only conclude that the state observer errors and tracking errors satisfying that $\|e_i\| \leq \max_{k \in M} \sqrt{2D/(C\lambda_{\min}(P_i^k))}$ and $|z_{i,1}| \leq \sqrt{2D/C}$, but do not converge to zero. However, according to [13]–[20], we can make both the state observer errors and tracking errors to be small by increasing the design parameters $l_{i,j}^k$, $\lambda_{\min}(Q_i^k)$, $c_{i,j}^k$, and $\delta_{i,j}^k$, or decreasing $\tau_{i,j}^k$ and $\eta_{i,a}^k > 0$, ($a = 1, 2, 3, 4$, $i = 1, \dots, N$; $j = 1, \dots, n_i$).

VI. SIMULATION EXAMPLE

In this section, a simulation example is given to illustrate the effectiveness of the proposed adaptive fuzzy decentralized output feedback control approach.

Example 1: Consider the following switched uncertain nonlinear large-scale systems:

$$\begin{cases} \dot{x}_{i,1} = x_{i,2} \\ \dot{x}_{i,2} = \frac{D_i^{\sigma(t)}(u_i^{\sigma(t)})}{J_i} + \left(\frac{m_{i,g}}{J_i} - \frac{Kr^2}{4J_i}\right) \sin(x_{i,1}) \\ \quad + \frac{kr(l-b)}{2J_i} + \Delta f_{i,2}^{\sigma(t)}(x_{i,2}) + \Delta_{i,2}^{\sigma(t)}(\bar{y}) \\ y_i = x_{i,1}, \quad i = 1, 2 \end{cases} \quad (58)$$

where the switching signal $\sigma(t) = k \in \{1, 2\}$, $\Delta f_{1,2}^1(x_{1,1}, x_{1,2}) = x_{1,1} \sin(x_{1,1}x_{1,2})/J_1$, $\Delta f_{1,2}^2(x_{1,1}, x_{1,2}) = x_{1,2} \cos(x_{1,1})/J_1$, $\Delta_{1,2}^1(\bar{y}) = Kr^2/4J_1 \sin(x_{2,1})$, $\Delta_{1,2}^2(\bar{y}) = Kr^2/4J_1 \sin(x_{1,1}x_{2,1})$, $\Delta f_{2,2}^1(x_{2,1}, x_{2,2}) = x_{2,1} \sin(x_{2,1}x_{2,2})/J_2$, $\Delta f_{2,2}^2(x_{2,1}, x_{2,2}) = x_{2,2} \cos(x_{2,1})/J_2$, $\Delta_{2,2}^1(\bar{y}) = Kr^2/4J_2 \sin(x_{1,1})$, and $\Delta_{2,2}^2(\bar{y}) = Kr^2/4J_2 \sin(x_{1,1}x_{2,1})$.

Besides, the parameters in (51) are given as: $m_1 = 0.2$ kg and $m_2 = 0.25$ kg are the pendulum end masses, $J_1 = 0.05$ kg and $J_2 = 0.0625$ kg are the moments of inertia, $K = 10$ N/m is the spring constant of the connecting spring, $r = 0.5$ m is the pendulum height, $l = 0.5$ m is the natural length of the spring, and $g = 9.81$ m/s² is gravitational acceleration. The distance between the pendulum hinges is defined as $b = 0.4$ m. In this example $b < l$, which means that the pendulums repel one another when both are in the upright position.

In this paper, the tracking signals are chosen as $y_{1,r}(t) = \sin(0.4t)$ and $y_{2,r}(t) = \sin(0.4t)$.

Remark 4: Note that if the system (58) does not contain the switching signal $\sigma(t)$ and $\Delta f_{i,j}^k(x_{i,j}) = 0$, it turns into a nonlinear large-scale system, which were extensively studied in [36]–[38].

The parameters in the dead-zone model (2) are selected as $m_{1,r}^1 = 2$, $m_{1,l}^1 = 1.3$, $d_{1,r}^1 = 0.4$, $d_{1,l}^1 = 0.6$, $m_{2,r}^2 = 1.1$, $m_{2,l}^2 = 2.2$, $d_{2,r}^2 = 0.2$, $d_{2,l}^2 = 0.5$, $i = 1, 2$.

Choosing the membership functions as

$$\begin{aligned} \mu_{F_{i,1}^k}(\hat{x}_{i,1}) &= \exp\left[-\frac{(\hat{x}_{i,1} - 3 + l)^2}{2}\right] \\ \mu_{F_{i,2}^k}(\hat{x}_{i,2}) &= \exp\left[-\frac{(\hat{x}_{i,1} - 3 + l)^2}{2}\right] \times \exp\left[-\frac{(\hat{x}_{i,2} - 3 + l)^2}{2}\right] \\ i &= 1, 2, \quad l = 1, \dots, 5, \quad k \in \{1, 2\}. \end{aligned}$$

The FLSs can be expressed in the form

$$\begin{aligned} \hat{f}_{i,j}^k(\hat{x}_{i,j} | \theta_{i,j}^k) &= (\theta_{i,j}^k)^T \varphi_{i,j}^k(\hat{x}_{i,j}) \\ i &= 1, 2, \quad j = 1, 2, \quad k \in \{1, 2\}. \end{aligned} \quad (59)$$

Then, we can construct the switched fuzzy decentralized state observer (16) as follows:

$$\begin{aligned} \dot{\hat{x}}_{i,1} &= \hat{x}_{i,2} + \hat{f}_{i,1}^k(\hat{x}_{i,1} | \theta_{i,1}^k) + l_{i,1}^k(x_{i,1} - \hat{x}_{i,1}) \\ \dot{\hat{x}}_{i,2} &= D_i^k(u_i^k) + \hat{f}_{i,2}^k(\hat{x}_{i,2} | \theta_{i,2}^k) + l_{i,2}^k(x_{i,1} - \hat{x}_{i,1}) \\ \hat{y}_i &= \hat{x}_{i,1}, \quad i = 1, 2 \end{aligned} \quad (60)$$

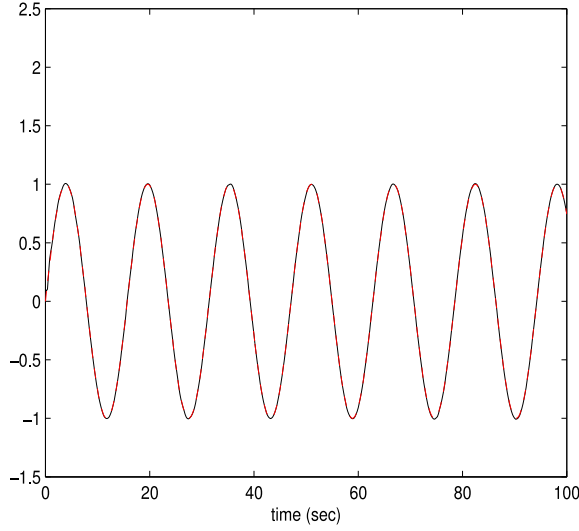
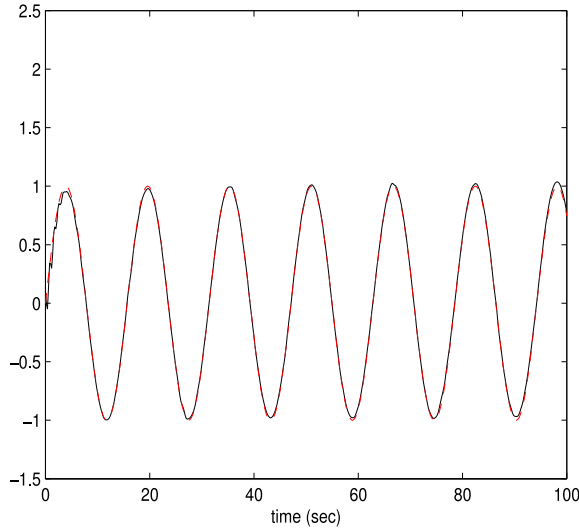
where $l_{1,1}^1 = 19$, $l_{1,1}^2 = 24$, $l_{2,1}^1 = 23$, $l_{2,1}^2 = 24$, and $l_{1,2}^1 = l_{2,2}^1 = l_{2,2}^2 = 25$.

In addition, in (19), by select $Q_1^k = Q_2^k = 6I$, we can obtain positive-definite symmetric matrices $P_1^k = \begin{bmatrix} 48.0 & -3.0 \\ -3.0 & 3.4 \end{bmatrix}$,

$$P_2^k = \begin{bmatrix} 27 & -3.0 \\ -3.0 & 1.94 \end{bmatrix}, \quad k \in 1, 2.$$

According to the intermediate function, the controller and parameter adaptive laws designed in the last section, u_i^k , $\alpha_{i,1}^k$, $\theta_{i,1}^k$, $\theta_{i,2}^k$, $\hat{m}_{i,r}^k$, $\hat{m}_{i,l}^k$, $\hat{d}_{i,r,m}^k$, and $\hat{d}_{i,l,m}^k$, $k \in 1, 2$; are chosen as

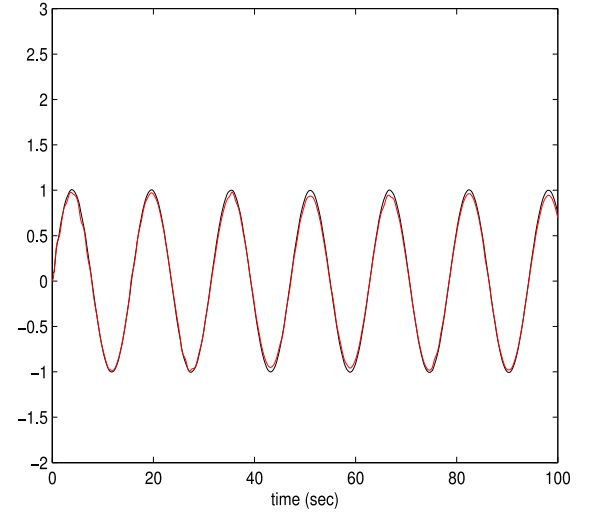
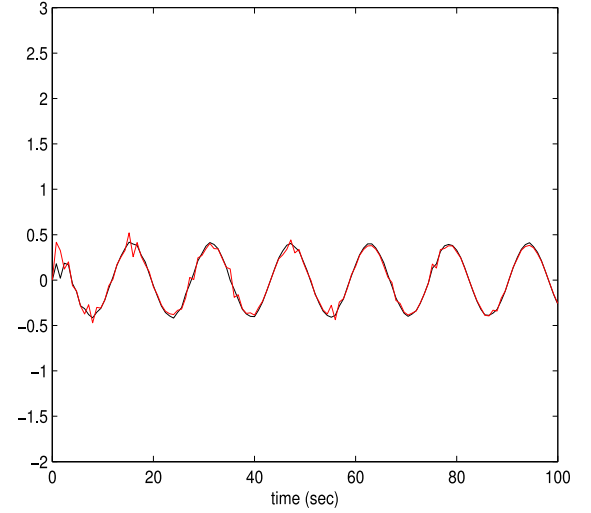
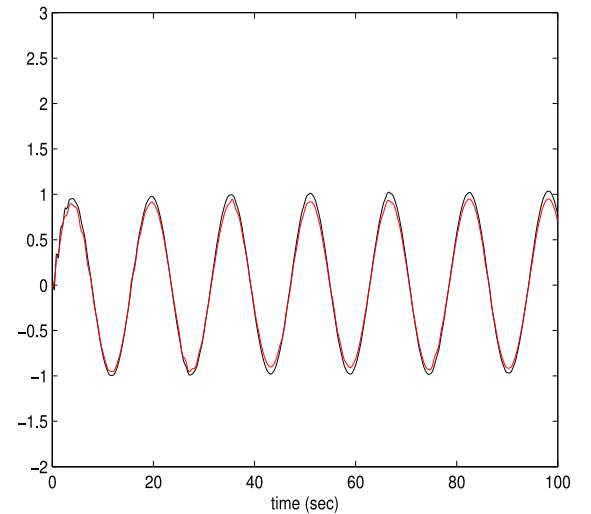
$$\begin{aligned} \alpha_{i,1}^k &= -0.1z_{i,1} - 6z_{i,1} - \theta_{i,1}^{kT} \varphi_{i,1}^k(\hat{x}_{i,1}) - 0.4 \cos(0.4t) \\ \dot{\theta}_{i,1}^k &= 0.4\varphi_{i,1}^k(\hat{x}_{i,1})z_{i,1} - 0.4\theta_{i,1}^k \\ u_{i,d}^k &= -0.1z_{i,2} - z_{i,2} - 2\left(\frac{\partial \alpha_{i,1}^k}{\partial x_{i,1}}\right)^2 z_{i,2} - \Xi_{i,2}^k \\ \dot{\theta}_{i,2}^k &= 0.4z_{i,2}\varphi_{i,2}^k(\hat{x}_{i,2}) - 0.4\theta_{i,2}^k \\ \dot{\hat{d}}_{i,r,m}^k &= -z_{i,n_i}0.02\delta_i^k + 0.01\hat{d}_{i,r,m}^k \\ \hat{m}_{i,r}^k &= \frac{u_{i,d}^k + \hat{d}_{i,r,m}^k}{\hat{m}_{i,r}^k} z_{i,n_i}0.02\delta_i^k + 0.01\hat{m}_{i,r}^k \\ \dot{\hat{d}}_{i,l,m}^k &= -z_{i,n_i}0.02(1 - \delta_i^k) + 0.01\hat{d}_{i,l,m}^k \\ \hat{m}_{i,l}^k &= \frac{u_{i,d}^k + \hat{d}_{i,l,m}^k}{\hat{m}_{i,l}^k} z_{i,n_i}0.02(1 - \delta_i^k) + 0.01\hat{m}_{i,l}^k. \end{aligned}$$

Fig. 1. Trajectories of y_1 (black line) and $y_{1,r}$ (red line).Fig. 2. Trajectories of y_2 (black line) and $y_{2,r}$ (red line).

The initial conditions are chosen as

$$\begin{aligned}
 x_{1,1}(0) &= x_{1,2}(0) = x_{2,1}(0) = x_{2,2}(0) = 0.1 \\
 \hat{x}_{1,1}(0) &= \hat{x}_{1,2}(0) = \hat{x}_{2,1}(0) = \hat{x}_{2,2}(0) = 0, \theta_{1,1}^1(0) = \theta_{2,1}^1(0) \\
 &= [0.1, 0.3, 0.5, 0.7, 0.9], \theta_{1,1}^2(0) = \theta_{2,1}^2(0) \\
 &= [-0.1, -0.3, -0.5, -0.7, -0.9] \\
 \hat{m}_{i,l}^k(0) &= \hat{m}_{i,r}^k(0) = \hat{d}_{i,l,m}^k(0) = \hat{d}_{i,r,m}^k(0) = 1 \\
 &(i = 1, 2; k \in \{1, 2\}).
 \end{aligned}$$

The average time τ_a is chosen as $\tau_a = 10$, and let $\mu = 2.5$ then, we can obtain $\tau_a = 10 > \ln 2.5/0.1$. Thus, the adaptive fuzzy decentralized output-feedback control problem of the resulting closed-loop system (58) is solvable under every switching signal $k = 1, 2$. The simulation results are shown in Figs. 1–9, where Figs. 1 and 2 express the trajectories of the system output y_i and tracking signal $y_{i,r}$ ($i = 1, 2$). Figs. 3 and 4 show the trajectories of $x_{1,j}$ and their estimations $\hat{x}_{1,j}$ ($j = 1, 2$), respectively. Figs. 5 and 6 show the trajectories

Fig. 3. Trajectories of $x_{1,1}$ (black line) and $\hat{x}_{1,1}$ (red line).Fig. 4. Trajectories of $x_{1,2}$ (black line) and $\hat{x}_{1,2}$ (red line).Fig. 5. Trajectories of $x_{2,1}$ (black line) and $\hat{x}_{2,1}$ (red line).

of $x_{2,j}$ and their estimations $\hat{x}_{2,j}$ ($j = 1, 2$), respectively. Figs. 7 and 8 show the trajectories of u_i^k ($i = 1, 2; k = 1, 2$), and Fig. 9 is the switching signal $\sigma(t)$.

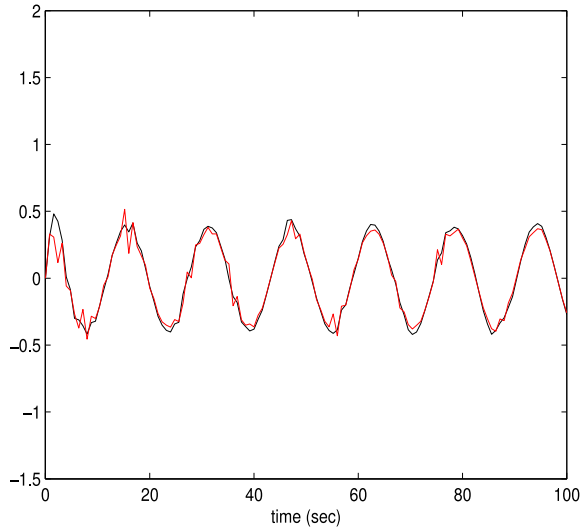


Fig. 6. Trajectories of $x_{2,2}$ (black line) and $\hat{x}_{2,2}$ (red line).

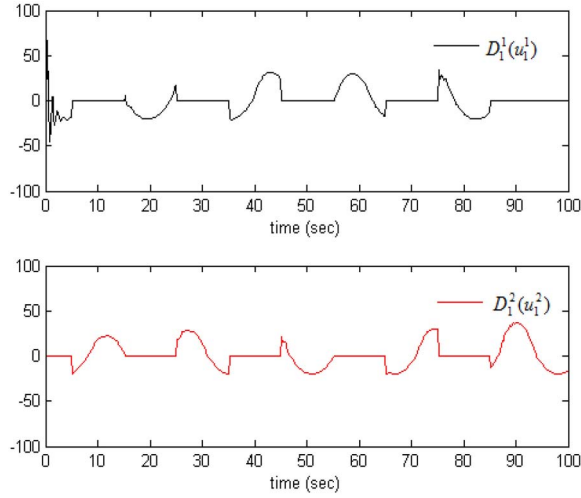


Fig. 7. Trajectories of u_1^k , $k = 1, 2$.

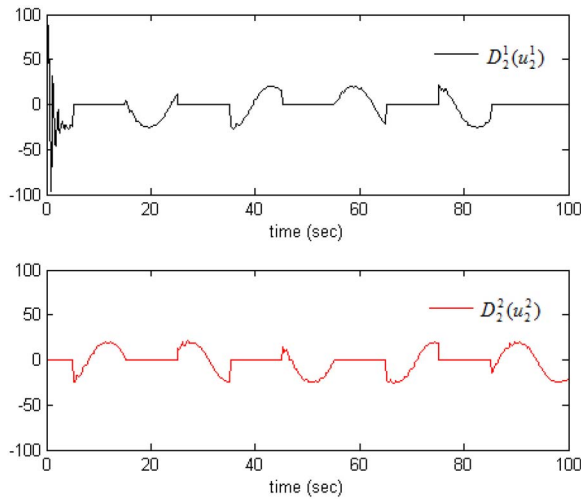


Fig. 8. Trajectories of u_2^k , $k = 1, 2$.

From the above simulation results, it is clear that although the switched nonlinear large-scale systems under study contain the uncertain nonlinearities and unmeasurable states, the

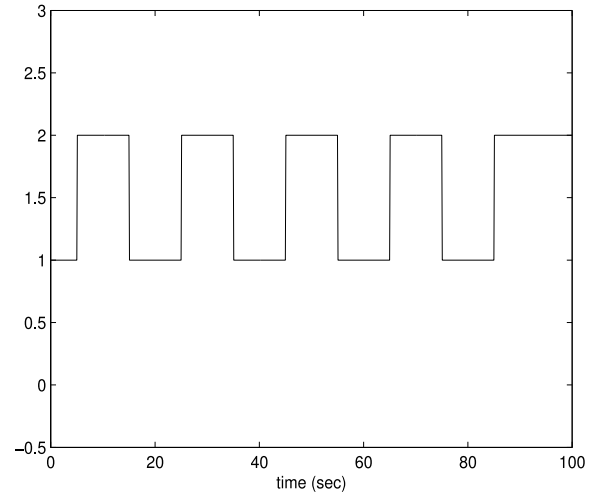


Fig. 9. Switching signals $\sigma(t)$.

proposed adaptive fuzzy decentralized output feedback controllers are able to guarantee the stability of the control systems and the tracking errors remain within a small neighborhood of the origin by appropriate choice of the design parameters.

VII. CONCLUSION

This paper has studied the adaptive fuzzy decentralized control design problem for a class of switched nonlinear large-scale systems in strict-feedback form. The considered switched systems have completely unknown nonlinear functions and dead zones, and without direct requirement of the states measurement. FLSs are utilized to model the switched nonlinear large-scale systems and a fuzzy decentralized state observer has been established for estimating the unmeasurable states. An adaptive fuzzy decentralized output feedback control scheme has been constructed by the backstepping design technique for each large-scale system. The stability of the whole switched control system has been proved via Lyapunov function and the average dwell-time method. Satisfactory tracking control performance has been achieved under the switching signals with average dwell time. The simulation results have demonstrated the effectiveness of the proposed adaptive fuzzy decentralized control method. Future research will be concentrated on adaptive fuzzy fault-tolerant control design for switched nonlinear systems with actuator faults based on the results of this paper.

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